

RoReEmo : Romania Reddit Emotion Dataset

Special Topics in Computational Linguistics

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Abstract

The following paper introduces the novel **ReRoEmo** (**R**eddit **R**omanian **E**motion) that contains romanian Reddit posts annotated with one out of five potential emotions: fear, anger, joy, sadness, neutral - inspired from **RED** (**R**omanian **E**motion **D**ataset) (Ciobotaru and Dinu, 2021) which contains tweets with the same potential labels. The dataset has been created as an aid in potential opinion mining tasks based on recent data. Moreover, it has been utilized as a potential extension of RED - creating a larger corpus of emotion annotated romanian short texts.

1 Introduction

As available online resources continued to grow, more documents capable of displaying individual opinions and ideas appeared and sparked interest for improving emotion detection systems and further analyzing individual opinion on diverse matters via comments, blogs or other forms of text data. This advancements enable new initiatives for low resource languages, such as Romanian, to be developed, although solutions are still scarce, compared to the ones available for English which can even explore emotion intensity. At the moment, there doesn't seem to exist an emotion labeled dataset on Reddit posts in romanian. RoRecEmo aims to bring diversity in the field of emotion detection in Romanian, either by itself, or combined with already existing methods such as the RED datasets. Moreover, RoRecEmo proposes new data that could capture relevant individual opinions, as only posts created after the 1st of January 2023 are selected.

2 Related Work

RED (Ciobotaru and Dinu, 2021) and RED v2 (Ciobotaru et al., 2022) have introduced the first supervised machine learning model for automatic Emotion Detection in Romanian short texts and its

enhanced next version designed for Multi-Label Emotion Detection. The article is highly based on the RED (Ciobotaru and Dinu, 2021) approach, with modifications at data preprocessing-level, experiments and hyperparameter values.

3 Dataset

3.1 Scraping process

The retrieved reddit posts have been obtained via keyword-based queries - which will be further referenced as query words - from some of the most active romanian subreddits, most related to casual conversation, questions, city-specific queries and trending topics that can also involve politics or news ("CasualRO", "Romania", "Cluj", "Bucuresti", "Timisoara", "Iasi", "Brasov", "Arad", "Oradea", "programare", "AskRo").

For each label, an initial selection of query words has been created out of:

- *synonyms* of each word defining the corresponding label extracted from two publically available online romanian dictionaries ([Dex-online](#)) **dexonline** and ([Sin0nime](#)) **Dictoo**
- *literals* (words that refer to the same concept) of each word defining the corresponding label from (Dumitrescu et al., 2018) RoWordNet

The final list of query words contains additional query words - valid words from the corresponding word families, obtained by adding suffixes / prefixes to the stem of the original words.

For each label, and for each query word in the initial list of selected words, Reddit posts have been selected based on how well they represent the emotion and whether they contain at least one word from the extended list of the query word.

Other implemented constraints include: a *post score* ≥ 1 , where *post score* is defined as *upvotes* - *downvotes* (users can upvote a post if

they find it valuable or downvote it if they consider it low-quality, off-topic, rule-breaking, or non-contributory content); a maximum character length of 800 (as the dataset focuses on short Romanian texts); and a creation date after January 1, 2023, since RoReEmo aims to capture recent trends in emotions within the top Romanian subreddits - thus, the selected timeframe is 1st of January 2023, 17th of February, 2025.

The **PRAW** (**P**ython **R**eddit **A**PI **W**rapper) library has been used to retrieve the posts.

3.2 Annotation process

Perhaps the major limitation of the dataset is the presence of a single annotator. This is addressed in the limitations section and the dataset has been made publicly available at and is open for relabelling. Upon completing the scraping process, for each label a .csv file has been obtained containing details about each post such as the subreddit it was extracted from, text, title, score, upvote ration, date it was created, id, label and the matched query words. The approach slightly differs from the one adapted for the RED dataset, the reason being the lack of posts that clearly expressed the specific emotion - an observation could involve the general aspect of posts, as a lot are formulated as questions; in contrast, post comments could better portray emotion as they offer tailored responses to the posed question or presented situation via the post. The selected posts have been manually checked - ambiguous posts have been dropped. For the *neutral* emotion, posts obtained with the other queries that were manually annotated were considered, alongside additional data scraped from another subreddit, *RoFiscalitate2* which mostly contains posts regarding generic financial questions.

Label	Query Words count	Posts count
fear	33	130
anger	29	147
joy	29	87
sadness	19	83
neutral	-	102

Table 1: Query words and posts count per label

3.3 Data processing

The following text preprocessing steps have been followed:

- removal of the hashtag sign #, while keeping

the following word which may be relevant

- removal of the zero-width space character, resulting from the scraping process
- replaced s and t cedilla-letters to comma-letters
- replaced email addressed with <EMAIL>, phone numbers with <TEL>, url with <URL> with the help of regex
- replaced proper nouns with <PROPN> with the help of *ro_core_news_lg* from spacy (Hon-nibal and Montani, 2017)

The last two steps have been replicated from (Ciobotaru et al., 2022). Punctuation, emojis and emoticons were kept as they are probably indicative to emotions. Upon preprocessing, the dataset has been shuffled and divided into train (80%), validation (10%) and test (10%).

4 Methods and results

4.1 Benchmark

Classic machine learning models have been utilized to set a benchmark, similarly to the approach presented in RED (Ciobotaru and Dinu, 2021). Of note is that experiments with different values for have been conducted, setting it to 10. For feature extraction, the same approach involving Tf-Idf has been implemented.

4.1.1 Multinomial Naive Bayes

Emotion	Precision	Recall	F1-score	Support
Bucurie	1.00	0.12	0.22	8
Furie	0.43	1.00	0.60	15
Frica	0.69	0.69	0.69	13
Tristete	1.00	0.25	0.40	8
Neutru	0.33	0.10	0.15	10
Accuracy			0.52	54
Macro avg	0.69	0.43	0.41	54
Weighted avg	0.64	0.52	0.45	54

Table 2: Multinomial Naive Bayes (test set)

Test Accuracy: 51.85%

4.1.2 Logistic Regression

Emotion	Precision	Recall	F1-score	Support
Bucurie	1.00	0.38	0.55	8
Furie	0.45	0.93	0.61	15
Frica	0.67	0.62	0.64	13
Tristete	0.67	0.25	0.36	8
Neutru	0.60	0.30	0.40	10
Accuracy			0.56	54
Macro avg	0.68	0.49	0.51	54
Weighted avg	0.64	0.56	0.53	54

Table 3: Logistic Regression (test set)

Test accuracy: 55.55%

4.1.3 SVC

Emotion	Precision	Recall	F1-score	Support
Bucurie	1.00	0.25	0.40	8
Furie	0.41	1.00	0.58	15
Frica	0.89	0.62	0.73	13
Tristete	0.67	0.25	0.36	8
Neutru	0.67	0.20	0.31	10
Accuracy			0.54	54
Macro avg	0.73	0.46	0.48	54
Weighted avg	0.70	0.54	0.51	54

Table 4: SVC (test set)

4.1.4 LinearSVC

Emotion	Precision	Recall	F1-score	Support
Bucurie	0.75	0.38	0.50	8
Furie	0.54	0.87	0.67	15
Frica	0.71	0.77	0.74	13
Tristete	0.50	0.50	0.50	8
Neutru	0.25	0.10	0.14	10
Accuracy			0.57	54
Macro avg	0.55	0.52	0.51	54
Weighted avg	0.55	0.57	0.54	54

Table 5: LinearSVC (test set)

Test Accuracy: 57.40%

4.1.5 RandomForestClassifier

Emotion	Precision	Recall	F1-score	Support
Bucurie	0.75	0.38	0.50	8
Furie	0.42	0.87	0.57	15
Frica	0.92	0.85	0.88	13
Tristete	0.50	0.12	0.20	8
Neutru	0.60	0.30	0.40	10
Accuracy			0.57	54
Macro avg	0.64	0.50	0.51	54
Weighted avg	0.63	0.57	0.55	54

Table 6: RandomForestClassifier (test set)

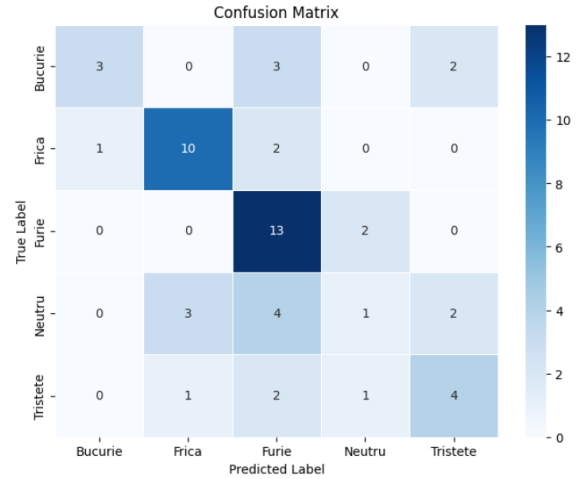


Figure 1: LinearSVC Confusion Matrix

Test Accuracy: 57.40%

4.2 BERT-based approach

BERT (Bidirectional Encoder Representations from Transformers) obtained state-of-the-art results in several natural language preprocessing tasks. The approach is similar to the one presented in RED (Ciobotaru and Dinu, 2021), however experiments with different values for batch_size, max_length (sequence) and the number of epochs have been conducted. The best results have been obtained for batch_size = 8, max_length = 512, epochs = 5. The texts are considerably larger than the ones presented in RED, thus a larger value for max sequence length was expected to perform better.

Emotion	Precision	Recall	F1-score	Support
Bucurie	0.80	1.00	0.89	8
Furie	0.83	0.77	0.80	13
Frica	0.75	0.60	0.67	15
Tristete	0.36	0.40	0.38	10
Neutru	0.62	0.71	0.67	7
Accuracy			0.68	53
Macro avg	0.67	0.70	0.68	53
Weighted avg	0.69	0.68	0.68	53

Table 7: Fine-tuned BERT (test)

Test accuracy: 67.92%

4.3 Results

The best results have been obtained by the BERT-based model (67.92%). For the classical models, LinearSVC performed the best (57.40%). The findings are similar to those of the RED dataset.

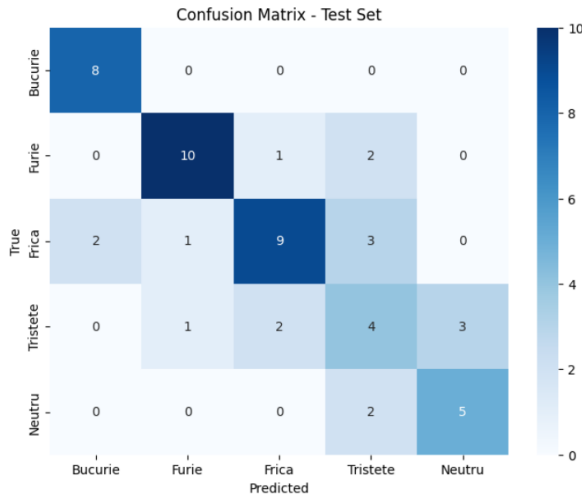


Figure 2: BERT Confusion Matrix

4.4 Augmenting RED

Augmenting the train dataset of RED with all of the samples from RoReEmo (train, validation, test) where 'RedditPostTitleText' corresponds to 'Tweet' and 'label' to 'Emotion' resulted in an increase in accuracy on the test set from 0.9086 to 0.9185.

Another observation is that only for the emotion of anger ("Furie") there is a decrease in metrics on the augmented dataset. As the RoReEmo dataset contains the most entries for this label and there has only been one annotator, there might be a few misinterpretations that affect the result.

Emotion	Precision	Recall	F1-score	Support
Bucurie	0.96	0.92	0.94	89
Furie	0.89	0.93	0.91	72
Frica	0.88	0.94	0.91	83
Tristete	0.92	0.86	0.89	81
Neutru	0.89	0.89	0.89	80
Macro avg	0.91	0.91	0.91	405
Weighted avg	0.91	0.91	0.91	405

Table 8: RED - (replicated)

Emotion	Precision	Recall	F1-score	Support
Bucurie	0.98	0.91	0.94	89
Furie	0.87	0.96	0.91	72
Frica	0.91	0.95	0.93	83
Tristete	0.92	0.88	0.90	81
Neutru	0.91	0.90	0.91	80
Macro avg	0.92	0.92	0.92	405
Weighted avg	0.92	0.92	0.92	405

Table 9: RED - augmented

5 Future Work

In terms of future work, the dataset could be further extended with multiple samples. Furthermore, a new version of the dataset could be created with additional emotions in a potential multilabel context.

6 Conclusion

In conclusion, a novel dataset has been introduced with up-to-date information collected from Reddit in Romanian and labeled with one of the five potential emotions: neutral, anger, sadness, joy, fear. The dataset is public and can be utilized for further research.

References

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A Appendix

Label	Query Words
fear	frică, groază, spaimă, sperietură, teamă, oroare
anger	furie, mânie, enervare, nervozitate, irascibilitate
joy	bucurie, fericire, veselie, încântare, mulțumire
sadness	tristețe, jale, regret, amărăciune, mâhnire

Table 10: Labels and initial list of selected query Words

Query word	Extended list of query words
frică	fricos, fricoasă, frici, înfricoșare, înfricoșat, înfricoșător, fricoși, fricoase, înfricoșată
teamă	temere, temeri, temut, teamăt, temută
groază	îngrozitor, groaznic, groaznică, îngrozitoare
spaimă	însăimântare, însăimântat, însăimântător, însăimântătoare
sperietură	speriat, speriată, speria, speriați, speriate

Table 11: Extended lists for query words corresponding to the label *fear*

Query word	Extended list of query words
furie	înfuriere, furios, furioasă, înfuriat, înfuriată, furioși, furioase
mânie	mânios, mânioasă, mânia, mânioși, mânioase
enervare	enervat, enervată, enervantă, enerva, enervați, enervate, enervant
nervozitate	nervos, nervoasă, nervi
irascibilitate	irascibil, irascibilă

Table 12: Extended lists for query words corresponding to the label *anger*

Query word	Extended list of query words
bucurie	bucuros, bucuroasă, bucurii, bucuroși, bucuroase
fericire	fericită, fericit, fericiti, fericite
veselie	înveselit, înveselită, înveseli, înveseliți, înveselite
încântare	încânta, încântată, încântat, încântați, încântate
mulțumire	mulțumi, mulțumit, mulțumită, mulțumiți, mulțumite

Table 13: Extended lists for query words corresponding to the label *joy*

Query word	Extended list of query words
tristețe	trist, tristă, triști, triste, întristare, întristat, întristată
regret	regrete, regreta, regretat, regretată
mâhnire	mâhni, mâhnit, mâhnită, mâhniți, mâhnite

Table 14: Extended lists for query words corresponding to the label *sadness*